Yunus Serhat Bıçakçı, PhD

Geospatial AI | Vision-Language Models | Remote Sensing | Spatial Data Science

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Summary

I am an Assistant Professor and geospatial AI researcher bringing together expertise in artificial intelligence and spatial data science to study urban and environmental questions. I founded and currently lead the GeoAI research group at Marmara University. Building on research at UCL and my continuing affiliation with the University of Glasgow, I work with five PhD students and four postdoctoral researchers across Marmara University and Glasgow to develop research ideas, joint funding applications and publications. I lead two funded projects as Principal Investigator: an Istanbul street-imagery project developed jointly with Glasgow colleagues, and SokakPuan, a platform for evaluating streets and public spaces. My research combines this collaborative approach with practical, open outputs, including research software and trained model releases. It has also received international recognition through research competition wins and a coauthored award-winning conference paper. My work at UCL also underpinned British Academy endorsement for the UK Government's Global Talent visa, recognising academic excellence and research promise. Beyond my university affiliations, my research partnerships extend to The Alan Turing Institute in the UK and DSO in Singapore.

My people-management skills are grounded in leading two funded projects and providing day-to-day research guidance to PhD students and postdoctoral colleagues. I help researchers refine research questions and develop project ideas, alongside supporting their wider academic development. Through the inzva Mentorship Program, I have mentored six undergraduates over three annual cohorts; several have gone on to master's and doctoral study, including at ETH Zurich and Carnegie Mellon University. I have also served on recruitment interview panels for five positions in the UK and three in Türkiye. I also contribute to research evaluation beyond academia, regularly reviewing technopark companies' projects. I also contribute to the wider research community by serving on conference committees and organising academic panels and workshops.

Academic and Research Appointments

2026–present **Assistant Professor**
Department of Artificial Intelligence and Machine Learning, Faculty of Applied Sciences, **Marmara University**, Istanbul, Türkiye.
Conduct research and teach in artificial intelligence and data science. Contribute to curriculum development, research projects and academic collaboration.

2024–present **Research Associate, then Research Software Engineering Scientist / Affiliate**
Geospatial Data Science Group, School of Geographical and Earth Sciences, **University of Glasgow**, UK. Staff page
Research Associate for six months on the Multi-LM project funded by The Alan Turing Institute; since then Research Software Engineering Scientist as an affiliate. Collaborative research, manuscript preparation, conference presentations and joint project development.

2024–2026 **Assistant Professor**
Vocational School of Social Sciences, **Marmara University**, Istanbul, Türkiye.
Taught undergraduate courses and contributed to research, curriculum development and departmental activities.

2023–2024 **Postdoctoral Researcher**
Jill Dando Institute of Security and Crime Science, **University College London**, UK.
Conducted postdoctoral research in geospatial analysis and data science as a TÜBİTAK 2219 Fellow, on research leave from Marmara University. Collaborated on publications and presented research at academic events.

2022–2023 **Assistant Professor**
Vocational School of Social Sciences, **Marmara University**, Istanbul, Türkiye.
Taught undergraduate courses and contributed to research, curriculum development and departmental activities.

2021–2022 **Research Assistant, PhD**
Vocational School of Social Sciences, **Marmara University**, Istanbul, Türkiye.
Supported teaching, research and departmental activities.

Academic Leadership and Professional Service

2026–present **Founder and Group Leader, GeoAI Research Group**
Marmara University, Istanbul, Türkiye. Group website

2026–present **International Communication Chair, GIScience 2027**
One of five International Communication Chairs, Fourteenth International Conference on Geographic Information Science, Tongji University, Shanghai. Organizing committee

2025–present **Project Reviewer, Marmara University Technology Transfer Office**
Regularly review projects for companies based in the technopark through the university's Technology Transfer Office.

2023–present **Recruitment Interview Panel Member, UK and Türkiye**
Served on recruitment interview panels for five distinct positions in the UK and three in Türkiye.

2023–present **Delegate, Environmental Criminology and Crime Analysis (ECCA)**

2022–2023 **Vice Head of Department, Marmara University**
Department of Office Technologies and Data Management, Vocational School of Social Sciences.

2022 **Editorial Board / Co-Editor**
The Social Research Journal of Marmara University.

2021–present **Journal Refereeing**
International Journal of Digital Earth, *PLOS ONE*, *International Journal of Environment and Geoinformatics*, *The Social Research Journal of Marmara University* and *International Humanities and Social Science Review*.

Research Funding and Fellowships

2026–2027 **AI-Based Analysis of Istanbul Street-Level Imagery and Creation of an Open Dataset**
Principal Investigator, TRY 546,500 awarded
Research Universities Support Program, project 12127, 12 months.
Lead a GeoAI project that uses AI models to turn about three million existing Mapillary images of Istanbul into an open dataset of street-level measures: segmentation results, perception estimates, descriptions and embeddings, with a manually labelled reference subset stratified by district, lighting and road type. Developed the successful application jointly with a Professor and a postdoctoral researcher at the University of Glasgow.

2026 **SokakPuan**
Principal Investigator, TRY 100,000 awarded
STEPS Open Innovation Call (Sokaklar Dönüşüyor), supported by EIT Urban Mobility.
Lead the development and municipal piloting of a platform that uses imagery, GIS, artificial intelligence and citizen feedback to evaluate streets and public spaces. Around schools that municipalities selected for street redesign, the team captured 360° imagery with a GoPro MAX 2 camera and published it on Mapillary, then built a baseline of street elements, such as crossings, bollards, trees and speed bumps, and of perceived safety; the analysis will be repeated after the redesign to measure change. One of six ventures selected for Sokaklar Dönüşüyor, alongside eight municipalities.

2023–2024 **TÜBİTAK 2219 International Postdoctoral Research Fellowship**
Fellowship recipient, EUR 25,200 awarded for 12 months, plus travel
Competitive TÜBİTAK fellowship supporting postdoctoral research at University College London on artificial intelligence, crime mapping and urban data science.

2024 **Multi-Lingual and Multi-Modal Location Information Extraction (Multi-LM)**
Research Associate for six months, total project funding GBP 394,911
The Alan Turing Institute, grant 317199-2.
Joined the project at Glasgow in 2024. Contributed to application development and geolocation research and coauthored a paper from the project.

Awards and Recognition

2026 **1st-Ranked Submission, PalmCity Test Task**
Ranked first in the leaderboard snapshot of 5 October 2026, with 57.08% mean IoU and 67.96% mean F1 on the hidden test split. Submission 961441. This is a dated leaderboard result, not a final competition award.

2026 **Best Short Paper Award, AGILE 2026**
Coauthor of the short paper on semantic neighbourhoods in London's Airbnb properties that received the award at the 29th AGILE Conference, Tartu, Estonia.

2025 **1st Place, WHISPERS ISASeg Challenge, Phase 1 and Phase 2**
First in both phases as an individual entry, using multimodal ensembles to map impervious surfaces from Sentinel-1 and Sentinel-2 imagery (Phase 1 and Phase 2 leaderboards). The resulting single-authored article was accepted by the *International Journal of Environment and Geoinformatics* in October 2026.

2024 **Global Talent Endorsement, The British Academy**
Received British Academy endorsement for the UK Government's Global Talent visa on the strength of research undertaken at UCL, recognising academic excellence and research promise.

Education

2014–2021 **PhD in Applied Informatics, Istanbul Technical University**, Türkiye
Geographical Information Technologies Program.
Dissertation. *Spatio-Temporal Analysis of Applied Electronic Monitoring on Parolees of Mobility: A Case Study in Istanbul.* Thesis
Analysed GNSS-tracked electronic-monitoring data for 77 people on probation in Istanbul (2014–2016), measuring when they entered areas and how long they stayed, with a grid design that protected anonymity; GIS, PostgreSQL/PostGIS and R.
Supervised by Prof. Dursun Zafer Şeker.

2011–2013 **MSc in Information Systems, Sakarya University**, Türkiye
Management Information Systems Program. Supervised by Prof. Orhan Torkul.

2007–2011 **BEng, Ankara University**, Türkiye

2007–2011 **BA in Business Administration, Anadolu University**, Türkiye

Languages

Turkish: native. English: fluent in speaking, reading and writing.

Peer-Reviewed Journal Articles

2026 **AlphaEarth Embeddings Match Conventional Earth Observation for Street-Level Greenery**
Yunus Serhat Bıçakçı.
Photogrammetric Engineering & Remote Sensing. Accepted September 2026; in press, proofs due November 2026. Code archive | GitHub

2026 **Impervious Surface Segmentation from Multisource Satellite Imagery Using Early-Fusion Deep Ensembles**
Yunus Serhat Bıçakçı.
International Journal of Environment and Geoinformatics. Accepted October 2026; in production.

2026 **Accuracy and Computational Cost of Detector Fusion for Tank Detection in Drone Imagery**
Yunus Serhat Bıçakçı.
Journal of Sustainable Development Issues. Accepted 2026; production to follow. Model weights

2026 **Performance Comparison of Multimodal Vision-Language Models in Classifying Turkish Dishes**
Yunus Serhat Bıçakçı.
Sakarya University Journal of Computer and Information Sciences, 9(1), 119–133. DOI

2025 **Comparative Assessment of CNN and Transformer U-Nets in Multiple Sclerosis Lesion Segmentation**
Beytullah Sarıca, **Yunus Serhat Bıçakçı**, Dursun Zafer Şeker.
International Journal of Imaging Systems and Technology, 35(4), e70146. DOI

2023 **ATTransUNet: Semantic Segmentation Model for Building Segmentation from Aerial Image and Laser Data**
Yunus Serhat Bıçakçı, Beytullah Sarıca.
Nordic Machine Intelligence, 2(3), 7–9. DOI

2020 **Location-Based Analyses for Electronic Monitoring of Parolees**
Yunus Serhat Bıçakçı, Dursun Zafer Şeker, Hande Demirel.
ISPRS International Journal of Geo-Information, 9(5), 296. DOI

2019 **The Relationship Between Attitudes Towards Sport and Life Satisfaction: A Study of Students at Marmara University's Vocational School of Social Sciences**
(*Spora Yönelik Tutum İle Yaşam Doyumu Arasındaki İlişki: Marmara Üniversitesi Sosyal Bilimler Meslek Yüksekokulu Öğrencileri Örneği*; in Turkish).
Çağdaş Caz, **Yunus Serhat Bıçakçı**, Fatma Firuzan Nakipoğlu.
Gümüşhane Üniversitesi Sosyal Bilimler Enstitüsü Elektronik Dergisi, 10(3), 484–491. Full text

2017 **Spatio-Temporal Analyses to Estimate Speed Information for Transport Forecasting and Scenario Testing**
Yunus Serhat Bıçakçı, Beytullah Sarıca, Mete Ercan Pakdil, Batuhan Yazırlı, Hande Demirel.
Fresenius Environmental Bulletin, 26(1), 100–106. Publication link

Peer-Reviewed Conference Papers and Proceedings

2026 **A Graph-Based Community Detection Approach for Identifying the Semantic Neighbourhoods within London's Airbnb Properties**
Joseph Shingleton, **Yunus Serhat Bıçakçı**, Yu Wang, Ana Basiri, Meiliu Wu.
AGILE: GIScience Series, 7, 41. Peer-reviewed short paper, 29th AGILE Conference, 16–19 June, Tartu, Estonia. DOI
Best Short Paper Award.

2026 **Mapping the Semantics of the Street: A VLM-Driven Geospatial Analysis in Fatih, Istanbul**
Yunus Serhat Bıçakçı, Joseph Shingleton, Yu Wang, Anahid Basiri.
Proceedings of the 1st International Conference on Geospatial Artificial Intelligence (GeoAI 2026), Oral Presentation Papers; published after peer review; oral presentation, 3–6 June, Ghent, Belgium. DOI

2026 **The Invisible City: Not All Streets Are Seen Equally**
Yu Wang, Joseph Shingleton, **Yunus Serhat Bıçakçı**, Anahid Basiri, James Ackland.
Proceedings of GeoAI 2026 (Oral Presentation Papers), published after peer review; oral presentation, 3–6 June, Ghent, Belgium. DOI

2026 **Emergent Spatio-Semantic Structure in Large Language Model Embedding Spaces**
Joseph Shingleton, **Yunus Serhat Bıçakçı**, Yu Wang, Anahid Basiri.
Proceedings of GeoAI 2026 (Oral Presentation Papers), published after peer review; oral presentation, 3–6 June, Ghent, Belgium. DOI

Book Chapters

2026 **Towards GeoAI Foundation Models: A Multimodal Learning Framework with Geospatial Intelligence**
Meiliu Wu, Qunying Huang, Anahid Basiri, **Yunus Serhat Bıçakçı**, Joseph Shingleton.
In *Geography According to Foundation Models*, Frontiers in Artificial Intelligence and Applications, vol. 422, IOS Press, pp. 28–50. DOI

2025 **Big Data and Social Media Analytics: Opportunities for Interactive Crime Mapping**
Yunus Serhat Bıçakçı, Alina Ristea, Kate Bowers.
In M. Leitner (ed.), *New Research in Crime Modeling and Mapping Using Geospatial Technologies*, Geotechnologies and the Environment, Springer, pp. 3–25. DOI

2025 **The Use of Artificial Intelligence in Street-Level Image Analysis**
(*Yapay Zekanın Sokak Seviyesi Görüntü Analizinde Kullanımı*; in Turkish).
Yunus Serhat Bıçakçı.
In M. Tokmak (ed.), *Management Information Systems in Türkiye and Worldwide* (*Türkiye ve Dünyada Yönetim Bilişim Sistemleri*), YAZ Yayınları, Afyonkarahisar, pp. 77–93. Chapter | Book archive

2022 **Artificial Intelligence Applications in Tourism and Travel Agencies**
(*Turizm ve Seyahat Bürolarında Yapay Zeka Uygulamaları*; in Turkish).
Yunus Serhat Bıçakçı, Bora Aksu.
In D. Semiz Çelik (ed.), *The Impact of Digital Transformation on Tourism* (*Dijital Dönüşümün Turizme Yansımaları*), Beta Yayınları, Istanbul.

Research Software, Open Data and Interactive Applications

2026 **PalmCity Challenge: Panoramic Street Segmentation**
Yunus Serhat Bıçakçı. Code and experiment records | Model weights
Released training and inference code and three EoMT–DINOv3 ViT-L seed checkpoints for 32-class panoramic segmentation. The selected probability ensemble uses three scales and horizontal reflection; validation mean IoU was 60.95% and hidden-test mean IoU was 57.08%.

2026 **FireRisk: SigLIP2 for Aerial Fire Risk Classification**
Yunus Serhat Bıçakçı. Code | Full checkpoint | Frozen checkpoint
Released training code and checkpoints comparing full SigLIP2 fine-tuning with a frozen encoder for seven fire hazard classes. Associated arXiv preprint: DOI.

Interactive AI and Geospatial Applications
Developed Gradio and Streamlit applications on Hugging Face Spaces, including Location Predictor and Crime-Map. Practical front-end and back-end development combining user-facing interfaces, Python application logic, model inference pipelines, APIs and geospatial data processing in end-to-end research prototypes.

2026 **GeoAI-VLM: Geospatial Vision-Language Model Analysis**
Yunus Serhat Bıçakçı. DOI | GitHub | PyPI
Open source Python research software for Mapillary image collection, structured vision-language descriptions and geographic outputs, with embedding, search and clustering modules. Automated tests run on every change, and releases are archived on Zenodo.

2026 **Fatih Mapillary Street-Level Images (open dataset)**
Yunus Serhat Bıçakçı. DOI | Hugging Face
About 124,000 street-level images of Fatih, Istanbul, with linked layers of vision-language descriptions, ZenSVI perception and glare estimates, and Cityscapes semantic segmentation. Released in connection with the Istanbul street-imagery project. CC BY-SA 4.0.

2026 **sppt: Spatial Point Pattern Test for Aggregated Data (Python)**
Yunus Serhat Bıçakçı. DOI | GitHub | PyPI
Python port of Martin A. Andresen's R package *sppt.aggregated.data*, an area-based test used in crime-and-place research to compare the spatial distributions of aggregated counts. Distributed on PyPI with a Colab quick-start notebook; implements sparse-matrix-accelerated bootstrap resampling, the S-Index and robust S-Index, bivariate change detection, choropleth maps and GIS exports.

Research Methods and Technical Expertise

Geospatial analysis and data. GIS and spatio-temporal analysis; street-level imagery (Mapillary, Google Street View) and 360° panoramas; Earth observation and learned embeddings (Sentinel-1, Sentinel-2, AlphaEarth), including fusion of radar and optical imagery; environmental mapping; QGIS, ArcGIS, GeoPandas, PostgreSQL and PostGIS.

Computer vision. Semantic and panoramic segmentation (EoMT with DINOv3 ViT-L backbones, Mask2Former, CNN and Transformer U-Nets), object detection (YOLO and DETR ensembles for UAV imagery, open-vocabulary detection of street elements), image classification and vision transformers.

Multimodal AI and vision-language models. Fine-tuning and evaluation of CLIP-family models, including SigLIP2; zero-shot benchmarking of open vision-language models (including Qwen3-VL, InternVL3, Gemma 3 and Aya Vision); structured vision-language description of images at scale; multimodal and language-model embeddings; geospatial foundation models; retrieval-augmented generation with multimodal large language models for image geolocation; dataset curation.

Statistical modelling and evaluation. Count regression (negative binomial, hurdle and hierarchical Bayesian negative binomial models); inter-rater reliability (intraclass correlation); pre-registered, spatially blocked validation with locked test sets; block-bootstrap intervals; spatial autocorrelation diagnostics (Moran's I); bootstrap-based spatial point pattern tests; explanation-integrity analysis with Shapley-based attribution; graph-based community detection and clustering.

Programming and AI software. Python, R, SQL and Bash; PyTorch, Hugging Face Transformers, vLLM, FAISS, LlamaIndex, LangGraph, NumPy, Pandas, OpenCV and DuckDB; model quantisation and serving.

Front-end and back-end development. Gradio and Streamlit interfaces, Hugging Face Spaces, Python application development, APIs, model inference pipelines and geospatial data processing for interactive AI research prototypes.

Reproducible research and computing. Git, Docker, CI/CD, automated testing, experiment tracking, high-performance computing, Google Cloud and Microsoft Azure.

Invited Talks, Training and Outreach

The Alan Turing Institute, UK
Delivered two invited sessions: a training session on high-performance computing (HPC) and a talk on the relationship between environmental criminology and artificial intelligence.

TensorFlow Basics, Google DSC Bandırma University, December 2022. Video
Intro to Machine Learning, Google DSC Işık University, November 2022. Video

Manuscripts Under Review and Revision

2026 **Can generative vision models measure micro-place environments? An ecometric assessment using recorded crime**
Yunus Serhat Bıçakçı.
Journal of Quantitative Criminology. Under peer review.

2026 **Evaluating Explanation Integrity in a Vision Language Model for Urban Perception**
Joseph Shingleton, **Yunus Serhat Bıçakçı**, Meiliu Wu, Yu Wang, Ana Basiri.
Transactions in GIS. Revised manuscript resubmitted following major revision and returned to peer review.

2026 **A due diligence protocol for open geospatial products derived by artificial intelligence applied to Fields of The World in Türkiye**
Yunus Serhat Bıçakçı.
Discover Geoscience. Minor revision. Code and reproducibility materials

2026 **How Well Do Open Street-Level Images Cover the City? Spatial Coverage and Data Gaps in Fatih, Istanbul**
Yunus Serhat Bıçakçı.
Turkish Journal of Geographic Information Systems. Under peer review. Code

Preprints and Submitted Manuscripts

2026 **Vision Transformer Ensembles for Panoramic Street Segmentation**
Yunus Serhat Bıçakçı.
arXiv:2610.06063 (v1, 5 October 2026). Code | Models
Documents nine candidate systems, three-seed confirmation of two candidates, and eleven inference variants for PalmCity. The selected EoMT–DINOv3 ensemble ranked first in the 5 October 2026 hidden-test leaderboard snapshot; validation selection dependence and rare-class errors are reported.

2026 **SigLIP2 for aerial fire risk classification**
Yunus Serhat Bıçakçı.
arXiv:2610.03689. DOI
Compares full SigLIP2 fine-tuning with a frozen encoder for seven aerial fire hazard classes; code and checkpoints are publicly released.

2026 **Hybrid CNN-Transformer Ensemble for Enhanced Tank Detection in Aerial Imagery**
Yunus Serhat Bıçakçı.
Research Square preprint. DOI

2025 **Street-Level Geolocalization Using Multimodal Large Language Models and Retrieval-Augmented Generation**
Yunus Serhat Bıçakçı, Joseph Shingleton, Anahid Basiri.
arXiv:2509.01341 [cs.CV]. DOI

Selected Conference Presentations and Abstracts

2025 **Digitalising Probation with Geospatial AI: Street-Level Human Perception in Istanbul's Fatih District**
Yunus Serhat Bıçakçı, Alina Ristea.
Turkish Probation System 20th Year International Congress, 1–3 December, Antalya, Türkiye. Conference summary.

2025 **Multimodal Semantic Segmentation for Impervious Surface Mapping**
Yunus Serhat Bıçakçı.
15th Workshop on Hyperspectral Image and Signal Processing: Evolution of Remote Sensing (WHISPERS), 12–14 November, Bellaterra, Barcelona, Spain.

2025 **From Video to Recipe: Leveraging Vision Language Models for Culinary Analysis**
Yunus Serhat Bıçakçı.
4th International Congress on Food Researches (ICONFOOD'25), 16–18 October, Sivas Cumhuriyet University, Sivas, Türkiye. Full paper published in the congress Full Texts Book. Code

2023 **Using Big Data, Artificial Intelligence and Cloud Technologies for Interactive Crime Mapping**
Yunus Serhat Bıçakçı, Alina Ristea, Kate Bowers.
Stockholm Criminology Symposium, 12–14 June, Stockholm, Sweden. Programme

2023 **Application of Natural Language Processing Techniques for Crime Detection**
Yunus Serhat Bıçakçı, Alina Ristea, Kate Bowers.
Environmental Criminology and Crime Analysis (ECCA), 14–16 June, Stockholm, Sweden. Programme

2022 **Artificial Intelligence Applications for Social Municipality: The Case of Chatbot**
Yunus Serhat Bıçakçı.
International VI. Civil Strategy Symposium, 8–9 December, Istanbul, Türkiye.

Teaching and Mentorship

2026–2027 **Marmara University, Department of Artificial Intelligence and Machine Learning**
Autumn 2026: Data Structures (in Python) and Career Planning. Also Office Software and Database Applications at the Vocational School of Social Sciences.

2025–2026 **Marmara University, Vocational School of Social Sciences**
Information and Communication Technologies (autumn and spring), Office Applications, Media Literacy and Management Information Systems.

2024–2025 **Marmara University, Vocational School of Social Sciences**
Information and Communication Technologies, Management Information Systems, Telephone Communication and Time Management.

2022–2023 **Marmara University, Vocational School of Social Sciences**
Office Applications, Computer I and Media Literacy.

2023–2026 **Mentor, inzva Mentorship Program**
Mentored six undergraduate students in their third or fourth year across three annual cohorts, with two mentees each year. Provided guidance on research planning, technical skills, academic development and careers in computer science and AI. Several mentees have gone on to master's and doctoral study, including at ETH Zurich and Carnegie Mellon University. Wrote about my mentoring experience in the Community Feedback section of the 2025/2026 programme report.